# GUILHERME HENRIQUE RISSON SILVA

Software Engineer Student | Backend | Python | FastAPI | LLM

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## ABOUT

Hello, I'm Guilherme. Software Engineer Student based in Maringá, Brazil. Software Engineering student (Uninter) with a postgraduate degree in Data Science, building Python backends with FastAPI, data pipelines and LLMs. Currently working as a data analyst at Vitru Educação on UniCesumar academic planning systems.

**Work experience with Python / backend projects:**

- Built a teacher allocation optimizer (OR-Tools CP-SAT, FastAPI, React) that places ~200 faculty across 350+ classes per module in minutes — previously weeks in spreadsheets
- Developed a student-base forecasting model with under 2% error vs. realized
- Migrated planned-vs-actual audits from Power Query to Python on 1–40M+ row datasets
- Maintain the weekly curriculum-matrix pipeline for UniCesumar/Uniasselvi (6 layouts, 4 quality layers)
- Shipped Job Hunters: FastAPI/Next.js/PostgreSQL/LLM SaaS indexing ~98k jobs from 649 sources
- Open-sourced Teacher Allocation Optimizer (production system at Vitru); regulatory RAG eval MRR 0.361 → 0.644

**Core skills:** Python • FastAPI • Next.js • React • TypeScript • PostgreSQL • pgvector • DuckDB • Docker • OR-Tools • RAG • Gemini • pytest • GitHub Actions • Power BI • SQL • Pydantic

**Foundations:** Software engineering fundamentals, data systems, backend APIs, LLM evaluation and production data pipelines.

## EXPERIENCE

### Vitru Educação (UniCesumar) — Mid-level Data Analyst, Academic Planning

03/2026 – Present · Hybrid, Maringá, Brazil

- Teacher allocation optimizer (OR-Tools CP-SAT, FastAPI, React, TypeScript): ~200 faculty across 350+ classes per module in minutes — previously weeks in spreadsheets
- Student-base forecasting model: error under 2% vs. realized
- Planned-vs-actual audits in Python on 1–40M+ row datasets (migrated from Power Query)
- Weekly curriculum-matrix pipeline for UniCesumar/Uniasselvi: 6 layouts, 4 quality layers, auditable history
- Executive Power BI dashboards refreshed weekly for managers and leadership

### Linoforte Móveis — Assistant Operator

06/2022 – 12/2025 · Osvaldo Cruz, Brazil

- Production and quality KPI monitoring with corrective action on recurring stops
- Bottleneck mapping that reduced rework and waste on the line

## PROJECTS

- **Job Hunters:** Job SaaS with FastAPI, Next.js, PostgreSQL, and LLM. ~98k jobs · 649 sources.
- **Teacher Allocation Optimizer:** In production at Vitru. CP-SAT + FastAPI + React — now open source.
- **TikTok Viral Generator:** Agents that produce short pt-BR videos with a source on every fact.
- **Regulatory RAG Eval:** RAG over Brazilian Central Bank regulations. MRR 0.361 → 0.644.

## EDUCATION

- B.S. Software Engineering — Uninter — 10/2026 – 2030 (in progress)
- AI Software Engineer 4.0 (360 h) — Data Science Academy — 09/2026 – 12/2026
- Graduate certificate in Data Science (360 h) — UniCesumar — 03/2026 – 10/2026
- B.S. Civil Engineering — FAI — 2021 – 2025